FOLD AND FAULT EXERCISE

(1) Deposition of the sandstones

(2) Intrusion of the dolerite (intrudes sandstones and hence post-dates them)

Intrusion of a parallel-sided, dolerite sill, concordant with the bedding in the sandstones. The thickness of the sill has been affected by the subsequent folding, being 4½ m on the SE limb, 6½ m on the NW limb and 7 m in the hinge

(3) Folding (dolerite sill folded and hence folding post-dates it)

Folding of the sandstones/sill into an anticlinal antiform with a half-wavelength of about 150 m, fold axis 30/044, axial plane 032/70SE, average limbs 008/46E and 068/56NW, interlimb angle 96°. The bisector, 040/84SE has an apparent dip in the profile plane which is 17° steeper than that of the axial plane, indicating that the SE limb of the fold is thinner compared to the other limbs

(4) Faulting (NW-SE fault offsets fold structures and hence post-dates folding)

Formation of an oblique-slip fault, oriented 122/60SW, slip vector 24/287 and magnitude 56 m (50 m dextral strike-slip, 26 m normal dip-slip)

(5) Intrusion of the pegmatite (offsets NW-SE fault and hence post-dates it)

Intrusion of a 15½ m wide, parallel-sided, discordant, pegmatite dyke, 085/60S, with an opening direction (in map view) normal to dyke margins

(6) Faulting (NNW-SSE fault offsets pegmatite and hence post-dates it)

Formation of an oblique-slip fault, oriented 160/35E, slip vector 17/135 and magnitude 81 m (70 m dextral strike-slip, 40 m normal dip-slip)

(7) Uplift and erosion to the present topography